

Set Symmetric Difference

$$X = [\text{red}, \text{green}, \text{blue}]$$

$$Y = [\text{red}, \text{yellow}]$$

$$X \cap Y = [\text{red}]$$

$$X \oplus Y = [\text{green}, \text{blue}, \text{yellow}]$$

in X not in Y

or in Y not in X

$$(a+b)+c = a+(b+c)$$

$$x \in X \oplus Y$$

$$\Leftrightarrow (x \in X \text{ and } x \notin Y)$$

$$\text{or } (x \notin X \text{ and } x \in Y)$$

Exclusive -

$$\text{or } \Leftrightarrow (x \in X) \text{ Xor } (y \in Y)$$

$$x + y \bmod 2 \Leftrightarrow x \cdot \text{Xor} \cdot y$$

$$1+1=0$$

Addition is associative
well-defined mod 2

1971-1972

[1971-1972] = 2

[1971-1972] = 1

[1971-1972] = 1.5

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